NUTAN KUMAR NAIK

AI/ML Engineer · Geoscientist · Energy & Subsurface Data

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PROFESSIONAL PROFILE

Geoscientist and AI engineer with an M.Sc in Geology from IIT Kharagpur and doctoral research at NIT Rourkela. Lead AI Researcher at Algen AI and first author of two arXiv papers on AI observability and agent assessment. Trained in well-log interpretation at ONGC and subsurface exploration at CMPDI. Brings machine learning, LLM engineering and satellite remote sensing to subsurface, water and emissions problems in the energy and petrochemical industry.

PROFESSIONAL EXPERIENCE

Lead AI Researcher, Algen AI Private Limited, Bengaluru — 2025 – Present

- Lead research on Traccia, an OpenTelemetry-native observability, governance and compliance platform for AI agents and LLM applications.
- Co-designed its telemetry pipeline, semantic guardrail assessment and hashed trace ledger, which produces tamper-resistant compliance evidence for regulations such as the EU AI Act.
- Created AgentStripes, an evidence-graded maturity model for AI agents, and piloted it on 30 open-source agent repositories.

Doctoral Researcher, National Institute of Technology Rourkela — 2021 – Present

- Estimate estuarine water quality from Sentinel-2 satellite imagery with machine learning, calibrated on field measurements (Bhitarkanika, Odisha).
- Field sampling and laboratory analysis of water and sediment by ion chromatography, UV-Vis spectrophotometry and YSI ProDSS.
- Industry projects: mine-water chemistry for MCL (15 coal mines) and IMFA; slope stability for SAIL iron-ore mines; resistivity surveys for Indian Railways; rock mass rating of an underground coal mine.

INDUSTRIAL TRAINING

- **ONGC, Kolkata.** Well-log interpretation: lithology identification, multi-well horizon correlation, deltaic and deep-marine depositional systems.
- **CMPDI, Bhubaneswar.** Coal exploration: borehole geophysical logging, drilling, and 3D geological modelling in MINEX.

EDUCATION

Ph.D, Hydrogeochemistry & Remote Sensing, NIT Rourkela — 2021 – Present

M.Sc, Geology, IIT Kharagpur · CGPA 8.54/10 — 2020

B.Sc, Government Degree College, Sundargarh · 72.67% — 2018

PUBLICATIONS

- *Traccia: An OpenTelemetry-Based Governance Platform for AI Systems.* arXiv:2607.14309, 2026. First author.
- *AgentStripes: An Evidence-Graded Maturity Model for AI Agents.* arXiv:2610.18603, 2026. First author.

SKILLS

AI & data: Python, SQL, PyTorch, scikit-learn, XGBoost, SHAP, LLM APIs, OpenTelemetry, Docker, GCP

Geoscience: Well-log interpretation, sedimentology, structural geology, hydrogeochemistry, MINEX

Remote sensing & GIS: Sentinel-2, ESA SNAP, ENVI, ArcGIS, QGIS

Laboratory: Ion chromatography, UV-Vis spectrophotometry, YSI ProDSS, petrographic microscopy

ACHIEVEMENTS

IIT JAM 2018 AIR 108 · GATE 2020 AIR 234 · CSIR NET (LS) Dec 2023 AIR 40

Secretary, Technical Society, NIT Rourkela (2023–24)